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Economic analysis of small-scale poultry production in Kenyan medium-sized cities of Kisumu and Thika

Samuel Onyango Omondi

Lund University, Department of Human Geography
P.O Box 223 62 Lund, Sweden
University of Nairobi, Department of Agricultural Economics
P.O Box 29053-00625 Kangemi, Kenya
onyisam316@yahoo.com

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Abstract
Studies have shown conflicting results regarding the importance of urban agricultural production for household food security. This study, while recognizing the importance of urban agriculture on food security, focuses on the profitability and household income aspects of urban agriculture. This is achieved using a sample of indigenous chicken producers in the medium-sized cities of Kisumu and Thika, in Kenya. Urban indigenous chicken production mainly serves a dual role of food provision and income generation. The multivariate regression models show that access to high value markets, household income level, and the type of production system used significantly affect profitability of indigenous chicken farming. To obtain the desired welfare for smallholder poultry farmers, policies should be introduced to facilitate their access to high value markets. Such policies should also include provision of affordable high yielding poultry breeds, support for formation of farmer groups, and training farmers on feed production.
Key words: urban agriculture, poultry farming, profitability, Kenya

1. INTRODUCTION

Rapid population growth and uneven economic development in Kenya has rendered formal employment incapable of meeting job requirements of the population (KIPPRA 2016). With widespread poverty of 36 percent (2015/16) and an annual urbanization rate of four percent, poverty and food insecurity have increasingly become major problems in urban areas (Republic of Kenya 2011, 2012a; KNBS 2018). Urban households engage in different income generating activities in the formal and informal sectors, including businesses, urban agriculture, and petty trade among others (Omondi *et al*. 2017). The informal sector, which has been growing constantly, provides employment opportunities to a significant proportion of the population (KIPPRA 2016). In 2016, the informal sector provided 83 percent of total employment in Kenya, and grew at a much faster rate than the formal sector (KNBS 2017).

Urban-based agriculture, that is, the engagement of urban households in agriculture in either rural or urban areas, is an important livelihood strategy for many households in Sub-Saharan Africa (SSA) (Jayne *et al*. 2015). For instance, in Ghana, 43 percent of households in Techiman and Tamale practice farming in urban or rural areas (Ayerakwa 2017). The proportion of urban-based farmers is even higher in Thika and Kisumu, in Kenya (55%) and Copper Belt Province in Zambia (84%) (Smart *et al.* 2015; Omondi *et al*. 2017). Urban agriculture plays an important role of food provision and livelihood strategies for households in all income groups (Lee-Smith 2010). Income shares from urban agriculture in SSA range between 12 and 36 percent, thus contributing significantly to the household economy of the urban population (Zezza & Tasciotti 2010; Omondi *et al*. 2017). Therefore, given the high participation rates and significant income contribution, it is not surprising that development agencies, researchers, and (local) governments have focused much attention on urban agriculture (Mougeot, 2011).

The role played by urban agriculture in cushioning citizenry against world food price shocks and urbanization of poverty makes it an important policy issue (de Zeeuw & Dubbeling 2009; Zezza & Tasciotti 2010), particularly in countries where a significant proportion of the urban population practice agriculture. The unprecedented urban food demand caused by rapid urbanization, increase in urban population, and a dietary shift of increased consumption of animal products (Pingali, 2010), presents an opportunity for urban farming to enhance food security and reduce poverty among the practitioners. Urban farmers engage in agriculture to cater for their household food needs, either through direct consumption of their produce or by marketing part of the produce to earn an income (Omondi *et al*. 2017). Whereas cereal staples produced in urban spaces are mainly for household consumption, vegetables and livestock products are mainly for income generation (Prain & Lee-Smith 2010).

The decision to engage in urban livestock farming can therefore be assumed to be profit motivated, in addition to food provision and as a leisure activity. Urban poultry farms could be considered to be microenterprises, which combine factors of production and available technology to produce outputs. Microenterprises, including poultry production, trading, and service provision along the poultry value chain provide employment to a significant proportion of the population in developing countries (Khaleda 2013).

While a majority of studies analyze the role of urban agriculture in enhancing food security, using food security indicators (Zezza & Tasciotti 2010; Warren *et al*. 2015) or the availability of urban land for agriculture (Martellozo *et al*. 2014; Badami & Ramankutty 2015), this study focuses on urban agriculture from an economic perspective. It analyses the profitability of poultry farming and the factors that affect its profitability. This is important given that money earned from agricultural activities is directly linked to household welfare when channeled to household food needs and other expenses (Chege *et al*. 2015).

Urban poultry farming presents an important research topic and will probably remain relevant given the context of increasing demand for food by rising urban populations, mounting land constraints for more extensive types of urban agricultural production, and the dietary shift towards white meat consumption (Delgado *et al*. 1999; Kearney 2010; King'ori *et al.* 2010). Poultry production is an attractive business as it requires smaller space, less investment, and gives faster returns than large livestock (Omiti & Okuthe 2009). Additionally, poultry enterprises provide farming households with both animal protein and income.

In Kenya, several County governments have been and still continue to promote agro-based microenterprises as an alternative to formal employment. For example, as a part of plans to ensure food security, Kisumu County provided urban farming households with poultry production inputs such as Day-Old Chicks (DOCs) and feed[1]. This implies that the County government considers urban poultry farming as a viable economic activity. Even though all nodes of the poultry value chain, from inputs acquisition, production, inputs and outputs transportation, value addition, and marketing present opportunities for microenterprises, this study focuses only on indigenous chicken production-oriented microenterprises.

The aim of the study is to investigate the profitability of urban indigenous chicken production[2]. This is achieved by answering two research questions; What is the profitability of urban indigenous chicken production? What factors determine profitability of urban indigenous chicken production?

2. METHOD

a. Study sites and data

This case study is based on primary data collected during the months of July and October 2016. Data were collected at the household level, in medium-sized Kenyan cities of Kisumu and Thika. Thika is located in the Central part of Kenya, only 50 km North of Nairobi and has a population of 151 thousand inhabitants (Republic of Kenya 2012b). Kisumu is in the Western part of Kenya, with a population of 383 thousand (UN-HABITAT 2005; Republic of Kenya 2012b).

Using an urban agriculture baseline survey of 2013[3], a list of all poultry producing households in Thika and Kisumu was used as sampling frame. Households in the two cities were sampled randomly using MS-Excel random number generator, resulting in a total of 312 respondents, 177 in Kisumu and 135 in Thika. The sample consisted of households producing broiler, layer, and indigenous chicken among other poultry species. Indigenous chicken producers constituted the majority of sampled households, with 254 households producing indigenous chicken.

Data were collected on household demography, poultry production, management, and marketing. It is important to note that farmers were asked to refer to written accounts on production and marketing of poultry. However, most indigenous chicken farmers do not keep records on flock

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[1] This information is from interviews with agricultural and livestock officers in Kisumu in 2016.

[2] Even though the survey data showed that a variety of poultry are raised in the two cities, indigenous chicken was the most dominant.

[3] For a detailed description of the sampling procedure used in the baseline survey, see Omondi *et al*. (2017).

size, input use, and marketing. Most of the data was therefore based on farmers' recall. While this recall could be considered as a limitation, it is presently the only available source of such information. To ensure data quality and reliability, the figures provided were compared with figures from other sources. For example, input and output prices were compared with those reported by the County agricultural reports and were found to be within the range of those reported in the two cities.

In addition to summary statistics, descriptive comparative analyses across the two cities on costs, revenues, and gross margin were conducted. The study further applied multivariate regression analysis to analyze factors determining the profitability of indigenous chicken production. Some households had very few indigenous chickens, therefore, the sample was truncated based on the number of indigenous chicken kept and reason for keeping poultry, allowing a minimum of 20 birds produced for income generation. It was assumed that households producing 20 birds or more would not consume all the birds or eggs, especially if the main reason for production was income generation. This reduced the sample of indigenous chicken producing households from 254 to 157 for regression analysis.

b. Conceptual framework: determinants of farm profitability

Although urban farmers have varying objectives for farming, some consider it principally as a business, with an aim of making profit (Dongmo *et al*., 2010). Because of difficulties in measuring household utility, farm profit is often used as a proxy for welfare (Barrett *et al*., 2012). Net farm profit, which takes account of fixed costs, would have been the best indicator of farm profitability. However, difficulties in estimating the fixed costs such as housing costs, feeding and watering troughs, heating equipment, and accounting for depreciation costs limited the estimation to gross margin computation. The difficulties emanated from the fact that some of these costs were incurred long time ago, and coupled with a lack of records, made estimation of net profit difficult. Therefore, in this study, gross margin per bird has been used as a measure of farm profitability.

The dependent variable is the natural log of gross margin per bird. Gross margin was specified as $GM = TR - TVC$ where GM is the gross margin in Ksh. per bird and TR is the total revenue in Ksh., derived from sales of chickens, eggs, and manure. TVC is a summation of variable costs in Ksh., including costs of DOCs, feed, drugs, and heating. Most of the interviewed households used family labor for production of indigenous chicken. Labor was therefore excluded from calculation of gross margins. For multivariate analysis, gross margin was transformed to natural logarithm. For those households making losses, the loss was transformed by adding the absolute value of the highest loss plus one to make all values positive, after which they were transformed to natural logarithm (Bos & Koetter 2011).

c. Empirical model

The empirical model was specified as follows:

$$y = \beta_0 + \sum_{i=1}^{6} \beta_i x_i + \varepsilon$$

Where y is the dependent variable (gross margin-ksh./bird), β_0 is the constant, β_1 to β_2 are parameters to be estimated, x_i to x_6 are the explanatory variables, and ε is the error term. The choice of explanatory variables was informed by farm performance and agricultural marketing literature. The control variables used include access to market information, marketing channel, household location, land ownership, poultry production system, and gender of poultry enterprise owner (see appendix 1 on how these variables were measured). A set of multivariate linear regressions modeling the relationship between gross margin and independent variables were fitted. First, the explanatory variables were fitted individually to determine their power in determining the level of profit. All the variables were then included in the full model to assess their joint significance in determining the profitability of indigenous chicken production.

3. RESULTS AND DISCUSSION

Household demographic and indigenous chicken production characteristics

Table 1 presents household demographic characteristics of indigenous chicken producers in Kisumu and Thika. Farmers are on average in their mid-40s and have practiced poultry farming for close to seven years. However, poultry farmers in Thika are significantly older than their counterparts in Kisumu. A typical farmer has 11 years of formal schooling, meaning that they started high school but did not finish. On average, the farm households have close to five members, with Kisumu households being slightly bigger. These figures are higher than the average household size for the country (3.9) and urban areas (3.2) (Republic of Kenya 2015). This means that urban farming households are generally larger than the non-farming households (Lee-Smith 2010). The relatively larger size of urban farming households than the non-farming households could imply access to more family labor needed for poultry production. It could also imply that the households have more mouths to feed and, consequently a need to engage in urban agriculture as a source of food and income.

Table 1: Summary statistics of indigenous chicken farming in Thika and Kisumu, 2016

	Variables	Kisumu (n=105)	Thika (n=52)	Whole sample (N=157)
Household demographic characteristics [a]	Gender of household head (1=male, 0=female)	0.78	0.78	0.78
	Age of household head (years)	42.7 (1.2)	50.2 (1.8)	45.2 (1.0)[a]
	Experience (years)	7.0 (0.5)	6.7 (0.7)	6.9 (0.4)
	Education of household head [b] (years)	11.0 (0.4)	10.9 (0.3)	10.9 (0.3)
	Household size (number)	5.1 (0.2)	4.3 (0.2)	4.8 (0.2)[b]
	Distance to output market (km)	2.6 (0.2)	0.9 (0.27)	2.0 (0.2)
Gender of owner	Female owned	25.7	44.2	31.9
	Male owned	18.1	23.1	19.7
	Both male and female owned	56.2	32.7	48.4
Household income status [c]	Low-income	44.2	28.6	39.1
	Middle-income	34.6	32.7	34.0
	High-income	21.2	38.46	26.9
Production ,management, and marketing	Free-range	46.7	23.1	38.9
	Deep-litter	21.9	23.1	22.3
	Both free-range and deep-litter	31.4	53.8	38.9
	Supplementary feeding (1=yes, 0=otherwise)	0.90	0.92	0.91
	Bio-security (1=yes, 0=otherwise)	0.30	0.15	0.25[c]
	Keep records (1=yes, 0=otherwise)	0.18	0.17	0.18
	Access to market information (1=yes, 0=otherwise)	0.90	0.35	0.71[a]
	High value market	10.5	9.6	10.2
	Brokers/retailers	52.4	9.6	38.2
	Direct sales/neighbors	35.1	80.8	51.6
Land size	All households (acres)	1.4 (1.9)	0.2 (0.3)	1.0 (1.7)[c]
	Those with land only	1.9 (2.0)	0.3 (0.3)	1.3 (1.8)[c]

Note [a]: Age, experience, education, distance, number of birds, and gross margin represent the mean, the rest of the variables are proportions
Note [b]: During the time of the survey, education system in Kenya was as follows: Primary school education lasted eight years, secondary school education lasted four years while college/university lasted three or more years (Republic of Kenya 2015)
Note [c]: Low-income households earn less than Ksh. 15,000, medium-income households earn between Ksh. 15,001 and 30,000, high-income households earn above Ksh. 30,000 per month (1USD was equivalent to Ksh. 100 at the time of survey)
Note [d]: Figures in brackets are standard deviations, [a] Significant at 1%, [b] significant at 5%, [c] significant at 10%
Source: Author’s survey, 2016

Most of the households producing indigenous chicken enterprises earn low (39%) and middle-incomes (34%), and poultry farms are located approximately two kilometers away from output markets. In total, the majority of poultry enterprises are co-owned between male and female household members (usually husband and wife). However, in Thika, most of the poultry enterprises are female owned. The higher number of females than males engaged in poultry production in Thika is consistent with other urban agriculture studies reporting that females tend to dominate poultry production, while men keep larger livestock like pigs and dairy (Dongmo *et al*. 2010). The survey shows that 78 percent of the households own some land (not reported in the table) and respondents own on average about 1.34 acres of land, with households in Kisumu having significantly larger parcels than Thika (Table 1).

The free-range production system is the most dominant system because it is less costly because of reduced expenses on feed, while the deep-litter system has high management costs and is highly dependent on purchased feed (King'ori *et al*. 2010; Okello *et al*. 2010). The main problem with the free-range system is its vulnerability to disease outbreaks, mainly New Castle Disease (NCD) which causes high mortalities and can wipe out an entire flock (Omiti & Okuthe 2009; Okello *et al.* 2010).

Most farmers provide supplementary feeding, but only one out of four maintains bio-security measures. The majority of indigenous chicken producers only seek veterinary services when there are disease outbreaks (Omiti & Okuthe 2009; King'ori *et al*. 2010). In addition, only a small proportion of farmers (18%) keep poultry production and marketing records. Records are important for poultry farming in calculating returns from the enterprise (Ondwasy *et al*. 2006). Seven out of ten poultry farmers have access to market information. A majority sell their output in traditional markets, such as open-air markets and to neighbors, retailers, or brokers, and only ten percent are selling to high value markets (HVMs), including hotels and restaurants, butcheries, and processors. The predominant market channels in Kisumu are brokers and retailers while in Thika, most producers sell directly in markets or to neighbors (Table 1).

Share costs of production and profitability of indigenous chicken production

The cost of feed constitutes the largest share of total production costs for indigenous chicken, accounting for close to three quarters of total costs (Table 2). Indeed, some broiler and layer producers in Kenya have exited production because of the high cost of feed (Okello *et al.* 2010). The second largest cost is the purchase of DOCs or breeding stock. Costs for drugs, including vaccines, antibiotics, and veterinary services constitute the third largest cost. The distribution of production costs does not vary significantly between the two cities, except for costs of drugs which are slightly higher in Kisumu than Thika. The higher cost of drugs in Kisumu could be because most farmers in Kisumu use the free-range production system whose main constraint is disease outbreaks. Frequent purchases of drugs in Kisumu increases the share of drug costs relative to Thika.

Table 2: Shares of different costs in indigenous chicken farming in Kisumu and Thika, 2016

	Share of cost (%)		
Item	Kisumu	Thika	Whole sample
Feed	70	76	73
DOCs	18	22	19
Drugs	10[b]	1	7
Heat	2	1	1

[a] Significant at 1%, [b] denotes significant at 5%, [c] denotes significant at 10%
Source: Author's survey, 2016

Comparison of revenues, costs, and gross margins of indigenous chicken farming in the two cities show that they are all significantly higher in Thika than in Kisumu (Figure 1). Revenues are derived from sale of eggs, chicken, and manure. On average, each indigenous chicken farmer generates Ksh. 756 from each bird. However, farmers in Thika earn gross margins of Ksh. 1,185 per bird, which is more than double that of their Kisumu counterparts.

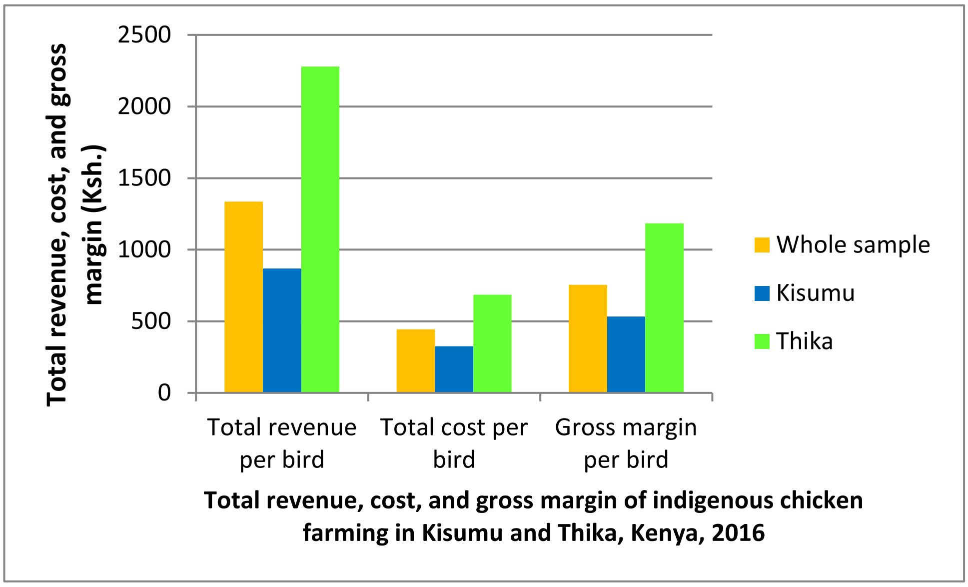

Figure 1: Total revenue, cost and gross margin of indigenous chicken production in Kisumu and Thika, 2016

Indigenous chicken production system, marketing, and profitability across different categories of farmers

The production systems and marketing with regards to profitability and variability in the two cities are summarized in Table 3. Simple comparison of means demonstrate that indigenous chicken farming is more profitable in Thika than Kisumu, and poultry enterprises co-owned by male and female household members appear to be more profitable than enterprises owned by males and females independently. Profitability also varies sharply across households based on their income status, particularly the middle-income households earn a significantly higher gross margin than the low and high-income households.

Table 3: Profitability of different categories of indigenous chicken farming and marketing in Kisumu and Thika, 2016

		Proportion (%)	Mean gross margin (Ksh./bird)
City	Kisumu	66.9	533
	Thika	33.1	1185[a]
Gender of owner	Female owned	31.9	598
	Male owned	19.7	693
	Co-owned by male and female	48.4	894
Income	Low-income	39.0	561
	Middle-income	34.0	1114
	High-income	27.0	583
Production system	Free-range	38.9	884
	Deep-litter	22.2	652
	Both free-range and deep-litter	38.9	692
Supplementary purchased feed	Yes	91.1	740
	No	8.9	958
Access to market information	Yes	71.3	736
	No	28.7	805
Market channel	High value market	10.2	899
	Brokers and retailers	38.2	511
	Direct sales/neighbors	51.6	919

[a] Significant at 1%, [b] significant at 5%, [c] significant at 10%
Source: Author's survey, 2016

Meanwhile, indigenous chicken farming under the free-range production system generates slightly higher gross margin than the deep-litter or the combination of free-range and deep-litter production systems. Consequently, those who provide supplementary purchased feed earn slightly less profits than those who do not. Surprisingly, although not statistically significant, farmers who had access to market information seem to make less profit than those who did not. Lastly, the majority of farmers who sell their chicken or eggs directly to neighbors or in markets, earn considerably more profits than those selling to brokers and retailers, and slightly more than those selling to HVMs (Table 2). Brokers often purchase poultry at the lowest price and later sell to other actors at a profit (Bett *et al.* 2012).

Indigenous chicken commercialization and consumption

Indigenous chicken producers in Kisumu and Thika have on average 68 birds. Overall, 58 percent of chickens and 24 percent of eggs produced are sold in the two cities. Indigenous chicken producing households consume 32 percent of chickens and 37 percent of eggs produced[4]. The household consumption level of eggs is much lower in these two cities than the average for Kenya, at 50 percent (Omiti and Okuthe, 2009). This could be an indication that urban poultry producers are more market oriented (Prain and Lee- Smith, 2010). As shown in Figure 2, the proportion of eggs and birds consumed and marketed vary significantly between the two cities. In Kisumu, a significantly higher proportion of indigenous chickens are marketed than in Thika. In contrast, a significantly higher proportion of birds are consumed by the producer households in Thika than in Kisumu. Likewise, a larger share of eggs is sold in Thika, while a higher proportion of chickens are consumed in Kisumu. A plausible explanation is that Kiambu County, in which Thika is one of the sub-counties, is the main producer of eggs in Kenya (Omiti & Okuthe 2009; Okello *et al.*

[4] It is important to note that the sum of proportions sold and consumed do not add up to unity because it is common practice for indigenous chicken farmers to retain some eggs and birds for reproduction (Okello *et al.* 2010).

2010). It is therefore easy for producers in Thika to market their eggs through the already established egg market in the County.

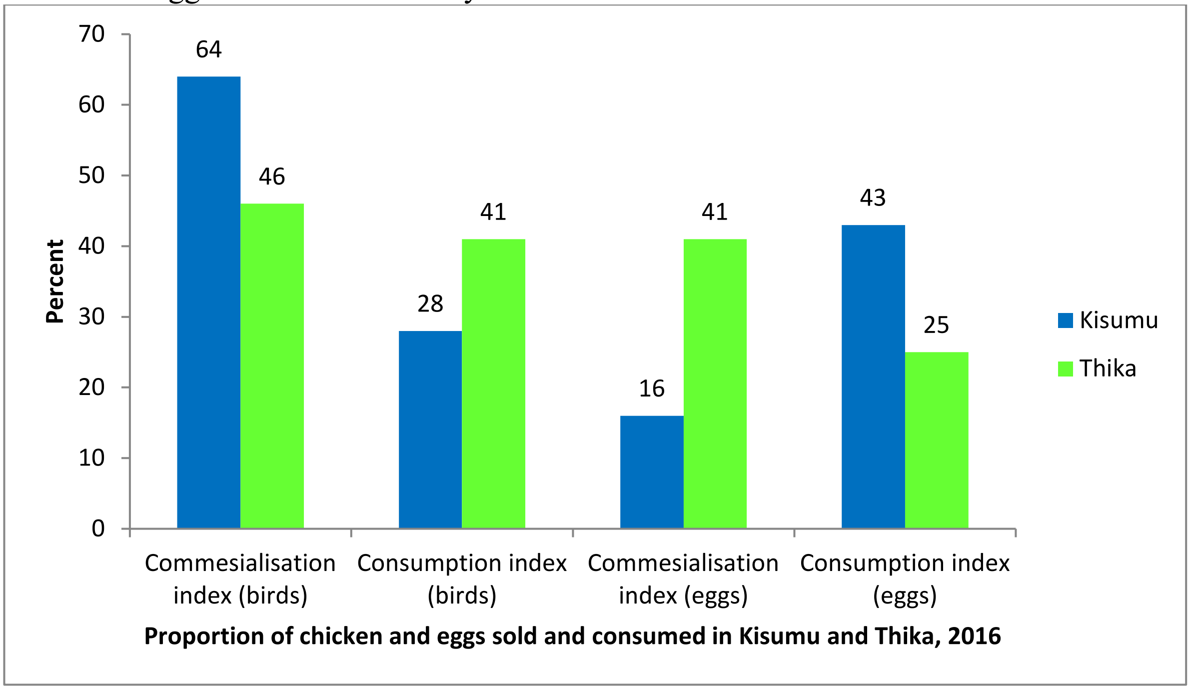

Figure 2: Proportion of chicken and eggs sold and consumed by urban poultry producing households in Kisumu and Thika, 2016
Source: Author's computation from 2016 survey

Indigenous chicken production constraints

Poultry diseases are the most common constraint reported by indigenous chicken producers (Figure 3). New Castle Disease (NCD) is the most serious disease afflicting indigenous chicken producers, particularly those using free-range production systems (Omiti & Okuthe 2009). Other important diseases include coccidiosis, fowl pox, and fowl typhoid. About a third of indigenous chicken farmers indicate that the high cost of inputs, especially feed, is a major challenge. Lack of market, low market prices for eggs and chicken, and pests (like fleas, lice, and mites) were reported as constraints by a minority of farmers. Most poultry diseases can be abated through vaccination and maintenance of hygiene. However, in Kenya, the vaccine for NCD, the most fatal poultry disease, is sold in doses of 100 birds and requires refrigeration for storage (King'ori *et al*., 2010). This hinders farmers with less than 100 birds and those who lack refrigeration facilities from buying the vaccine.

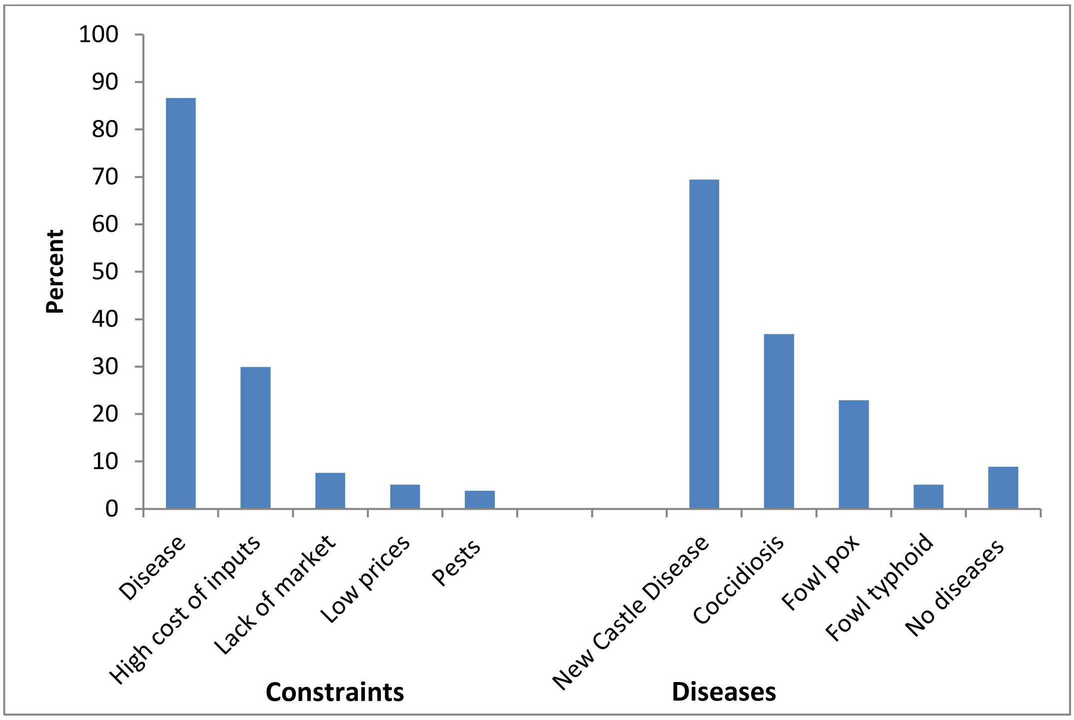

Figure 3: Constraints and diseases affecting indigenous chicken production in Kisumu and Thika as reported by sampled farmers, 2016
Note: Multiple responses allowed
Source: Author’s computation from 2016 survey

a. *Results for multivariate analysis*

Table 4 presents the relationship between gross margin and individual explanatory variables as well as the full model. Five out of six variables included in the full regression model are significant in determining indigenous chicken gross margin. In addition, the F statistic demonstrates that all the variables included in the model, considered jointly, are significant factors, at one percent level of significance in determining the amount of gross margin earned.

Table 4: Results of multivariate regressions of factors influencing indigenous chicken profitability

Variable	Coefficients (Robust standard errors)							
City	0.082 (0.409)							1.135 (0.504)[b]
Male owned		0.580 (0.506)						0.696 (0.479)
Co-ownership		0.626 (0.435)						0.571 (0.425)
Middle-income			0.884 (0.372)[b]					1.171 (0.355)[a]
High-income			-0.265 (0.522)					-0.259 (0.498)
Access to market information				0.539 (0.444)				0.812 (0.502)
Selling to HVM					1.109 (0.236)[a]			1.000 (0.365)[a]
Deep-litter system							-0.949 (0.507)[c]	-1.208 (0.519)[b]
Deep litter and free-range system							0.830 (0.397)[b]	-1.114 (0.284)[a]
Land size						0.181 (0.176)		0.164 (0.238)
Constant	5.669 (0208)[a]	5.292 (0.356)[a]	5.458 (0.299)[a]	5.317 (0.399)[a]	5.587 (0.200)[a]	5.600 (0.242)[a]	6.192 (0.213)[a]	4.450 (0.657)[a]
R^2	0.003		0.0465	0.0122	0.220	0.0024	0.0382	0.1708
p>F	0.8400		0.0132	0.2264	0.0000	0.3061	0.0427	0.0048

Notes: Figures in brackets are robust standard errors
[a] Significant at 1%, [b] denotes significant at 5%, [c] denotes significant at 10%
Source: Author's survey, 2016

Location of the farm and household income
The city where the farmer is located significantly influences the amount of gross margin generated as shown in the full regression model. Holding all other factors constant, indigenous chicken farming in Thika increases the profitability of the enterprise. This could be attributed to higher output prices and lower input prices in Thika than Kisumu because of a high concentration of feed millers in Thika (Omiti & Okuthe 2009). In addition, there is an already established egg market in Thika, which makes it easy for indigenous chicken producers in the region access the market (Nyaga 2008; Omiti & Okuthe 2009; Okello *et al.* 2010). However, considered individually, without controlling for the other explanatory variables, poultry production in Thika or Kisumu has no significant influence on indigenous chicken profitability. This implies that the bundle of advantages associated with being in a certain locality influences profitability of indigenous chicken production. For instance, cheaper feed and better output prices in Thika than Kisumu contribute to higher gross margin in Thika.

Middle-income households earn significantly higher gross margins than high-income households. This finding is consistent with summary statistics presented in Table 2. Poultry farming as an income diversification strategy appears to be more important to middle-income households than high and low-income households. Poultry farming is a viable part-time activity that diversifies household income (Molia *et al*., 2015). Even after controlling for other variables, household income level still influences profitability of chicken production.

Access to market information and HVMs

Access to market information provides an opportunity for farmers to sell their outputs to the market offering the best price. Access to market information and high expected prices provide incentives for smallholder farmers to enter commodity markets and increase supply (Olwande *et al*. 2015). Having access to market information alone has no significant effect on profitability of indigenous chicken productions. However, selling to HVMs is a significant determinant of indigenous chicken production, even after controlling for other explanatory variables. All other factors held constant, selling to HVMs significantly increases gross margin per chicken. Such markets generally offer higher and more stable prices than conventional markets (Neven, *et al.,* 2009). Furthermore, supermarkets demand higher volumes which reduce transaction costs (Neven, *et al.,* 2009).

Supplying to HVMs is contingent on consistency in the supply of the required volumes of quality products in a timely manner (Andersson *et al*. 2015; Ochieng *et al.* 2017). However, most indigenous chicken producers do not have contractual arrangements with buyers, creating uncertainty during production (Bett *et al.* 2012). Even though not statistically different, HVMs suppliers in the sample have 25 more birds than those selling to traditional markets. The problem of low volumes often excludes small producers from lucrative markets. Group marketing through producer groups, provision of extension services, advise to farmers on production and management practices, and adoption of improved technologies and high yielding poultry breeds would increase marketable surpluses and capacity to access HVMs (Neven *et al*. 2009; Ochieng *et al*. 2012; Olwande *et al.* 2015).

Production system

The type of production system a farmer utilizes affects their cost functions, and is a significant factor influencing profitability of indigenous chicken production. The amount of feed, extent of diseases control and management, and time devoted to poultry farming affects the performance of the enterprise. Both deep-litter and a combination of deep-litter and free-range systems significantly reduce profitability of indigenous chicken farming. This implies that producing indigenous chicken under free-range production system is more profitable than both deep-litter and a combination of deep-litter and free-range system. The cost of feed is, as mentioned, significantly lower in free-range system as birds scavenge for food (Okello *et al*. 2010).

The demand for indigenous chicken meat and eggs is rising in Africa (Molia *et al.* 2015). Undeniably, meat and eggs from indigenous chicken are gaining popularity in Kenyan urban centers because of their low saturated fats and cholesterol and the presumption that they are organically produced products (Omiti & Okuthe 2009; King'ori *et al*. 2010). Chicken meat fetches higher prices than other substitutes such as beef and pulses (Okello *et al*. 2010). Furthermore, indigenous chicken meat is more expensive than broiler meat (Okello *et al*. 2010; Bett *et al.* 2012). Despite this growing market, the full production potential of indigenous chicken has not yet been achieved in Kenya because of low survival rates and productivity (Ondwasy *et al.* 2006). The low productivity is because of poor performing breeds, low feed conversion efficiency, poor disease management, and low uptake of new technologies by farmers (King'ori *et al*. 2010). Indeed, disease control through vaccination and maintenance of hygiene in the free-range system could improve chicks' survival rate by 30 percent. Improved feeding, disease control, and proper housing would further improve survival rate to 80 percent (Ondwasy *et al*. 2006). The adoption of a full management package, comprising of feed supplementation, proper housing, proper brooding and chick rearing, and vaccination considerably increases productivity (Ochieng *et al*. 2010).

4. CONCLUSION/IMPLICATIONS FOR POLICY

Using the case of indigenous chicken farming in two medium-sized cities in Kenya, this study found poultry production to be a feasible and profitable venture. It contributes to the household food basket directly through consumption of chickens and eggs and indirectly through income generation. The findings of this study contribute to the urban agriculture debate by showing the extent to which an important activity such poultry production provides employment and income for a large share of urban dwellers in two medium-sized cities in Kenya.

Transformations in the food systems provide opportunities to smallholder farmers through increased income by selling to HVMs. Therefore, the findings of this study, support recommendations from other studies, that in order to improve the welfare of urban poultry producers, policies should be introduced to facilitate the linking of farmers to emerging HVMs (Neven *et al*., 2009; Chege *et al.,* 2015). Feed expenses constitute the largest share of costs in poultry production. In order to improve the performance of indigenous chicken production, cost of feed should be made more affordable. This could be achieved for example by County government agricultural and livestock departments training and supporting farmers on production of affordable high-quality poultry feed.

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Appendix 1: Measurement of variables used in regression models

Variable	Measurement
City	1=Thika, 0=otherwise
Male owned	1=poultry enterprise is male owned, 0=otherwise
Co-ownership	1= poultry enterprise is co-owned between a male and female household member, 0=otherwise
Middle-income	1=household belongs to a middle-income category, 0=otherwise
High-income	1= household belongs to a high-income category, 0=otherwise
Access to market information	1=household had access to market information, 0=otherwise
HVM	1=household sold produce to HVM, 0=otherwise
Deep-litter system	1=household produced chicken under deep-litter system, 0=otherwise
Deep-litter and free-range system	1=household produced chicken using a combination of deep-litter and free-range system, 0=otherwise
Land size	Acres